

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

B.Tech (Electronics and Telecommunication Engineering) SEMESTER - 4 Summer 2026 (Regular)

Course : Bachelor of Technology (Electronics and Telecommunication Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: 25AF1372PC401 - SIGNALS AND SYSTEMS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 A BIBO stable system satisfies:
- A) Unbounded input gives bounded output
 - B) Bounded input gives bounded output
 - C) Unbounded output always
 - D) Output equals input

- A causal system depends on:
- A) Future inputs
 - B) Past outputs only
 - C) Present and past inputs
 - D) Future outputs

- 3 An LTI system is completely characterized by its:
- A) Step response
 - B) Impulse response
 - C) Frequency response
 - D) Phase response

- 4 Which property states $xh=hx$?
- A) Associative
 - B) Distributive
 - C) Commutative
 - D) Linearity

- An LTI system is stable if:
- A) $h(t)$ is periodic
 - B) $h(t)$ is bounded
 - C) $\int |h(t)| dt$ is finite
 - D) $h(t)=0$

- Fourier series decomposes signals into:
- A) Exponentials
 - B) Polynomials
 - C) Impulses
 - D) Ramps

- 7 Parseval's theorem relates:
- A) Frequency and amplitude
 - B) Energy in time and frequency domains
 - C) Sampling and reconstruction
 - D) Phase and magnitude

- 8 Fourier coefficients of an odd function are:
 A) Cosine terms only
 B) Sine terms only
 C) Constant only
 D) Exponential only

- 9 Magnitude spectrum represents:
 A) Phase angle
 B) Amplitude versus frequency
 C) Time delay
 D) Energy only

- 10 Time shifting in time domain causes:
 A) Frequency scaling
 B) Phase shift in frequency domain
 C) Amplitude scaling
 D) Convolution

Laplace transform is an extension of:

- A) Fourier series
 B) Fourier transform
 C) Z-transform
 D) DFT

- 12 Initial value theorem is applicable to:
 A) Laplace transform
 B) Fourier series
 C) DTFT
 D) DFT

Q2. Solve the following.

- A) Describe the elementary signals used for testing systems such as impulse, step, ramp, exponential, sinc, and signum signals.
 B) State and prove sampling theorem with neat diagrams.

Q3. Solve the following.

- A) Solve the convolution of the following signal with graphical method:
 $x(n) = \{1 \ 2 \ 4 \ 5\}$, $-1 \leq n \leq 2$ $x(n) = \{1 \ 1 \ 1\}$, $0 \leq n \leq 2$
 B) Describe and define the properties of the system

Q4. Solve Any Two of the following.

- A) What is Fourier series? What are the Dirichlet conditions? State the formulas for trigonometric and exponential Fourier series.
 B) State and prove the any six properties of Fourier series.

Consider a Causal LTI system with frequency response $H(w) = \frac{1}{3+jw}$

For a particular input $x(t)$, this system is observed to produce the output, $y(t) = e^{-3t}u(t) - e^{-4t}u(t)$

Determine $x(t)$.

Q5. Solve Any Two of the following.

- A) Find the system frequency response $H(e^{jw})$ for
 $h(n) = 0.5\delta(n) + \delta(n-1) + 0.5\delta(n-2)$
 Also plot magnitude and phase response.

B) Find the CTFT of

1. $x(t) = e^{-at}u(t)$, $a > 0$
2. $x(t) = e^{at}u(-t)$, $a > 0$
3. $x(t) = \delta(t)$
4. $x(t) = \cos(\omega_0 t)$
5. $x(t) = \sin(\omega_0 t)$

C)

State and prove the property of differentiation in Z-domain. Find Z-transform of $g(n) = n\alpha^n u(n)$

Q6. Solve Any Two of the following.

A) State and prove the any six properties of Laplace Transform.

B) For the following system functions, check whether the corresponding LTI system is causal, anticausal, or non-causal by finding the inverse Z-Transform in each case.

$$1. H(z) = \frac{3-4z^{-1}}{1-3.5z^{-1}+1.5z^{-2}} \quad |z| > 3$$

$$2. H(z) = \frac{3-4z^{-1}}{1-3.5z^{-1}+1.5z^{-2}} \quad |z| < 0.5$$

$$3. H(z) = \frac{3-4z^{-1}}{1-3.5z^{-1}+1.5z^{-2}} \quad 0.5 < |z| < 3$$

Find the Inverse Laplace transform of

For a) $-2 < R(s) < 1$

b) $R(s) > 1$

$$H(s) = \frac{-3}{(s+2)(s-1)}$$

*** End ***